17 A train travelled along a track in 110 minutes, correct to the nearest 5 minutes.

Jake finds out that the track is 270 km long.
He assumes that the track has been measured correct to the nearest 10 km.

(a) Could the average speed of the train have been greater than 160 km/h?
You must show how you get your answer.

Distance: $265 \leq d < 275$ km.

Time: $107.5 \leq t < 112.5$ minutes.

1 mark
Use the correct bounds

Upper speed bound: largest $d \div$ smallest t.

Min	km
107.5	275
60	$\frac{6600}{43}$

$\times \frac{24}{43}$ $\times \frac{24}{43}$

1 mark
Use 275 and 107.5

1 mark
Convert to kilometres in 60 minutes

Speed $< 275 \div 107.5 \times 60 = 153.488\ldots$ km/h

1 mark
No. Even this upper bound is below 160 km/h.

(4)

Jake's assumption was wrong.
The track was measured correct to the nearest 5 km.

(b) Explain how this could affect your decision in part (a).

The upper distance bound falls to 272.5 km.

The time bounds stay the same, so the upper speed bound falls.

The answer remains No.

(1)

1 mark
A smaller upper distance gives a smaller upper speed

(Total for Question 17 is 5 marks)